

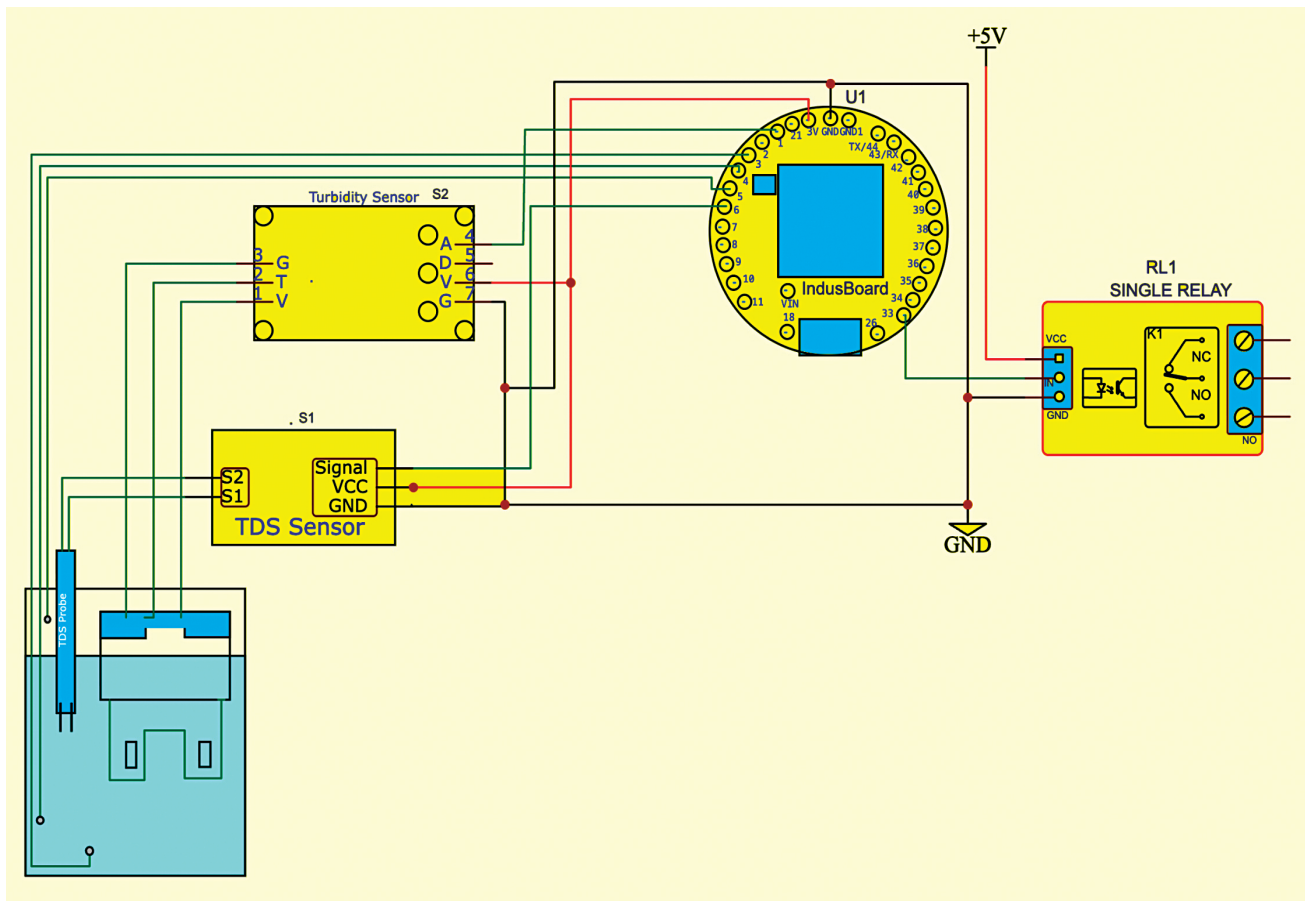


Fig. 2: Circuit diagram

```

Water_TDS.ino
1  #include <WiFi.h>
2  #include <WebServer.h>
3  #include <EEPROM.h>
4  #include "GravityTDS.h"
5
6  // WiFi credentials for STA mode
7  const char* sta_ssid = "JioFiber-4G";
8  const char* sta_password = "1234567890";
9
10 // Pin definitions for ESP32-S2
11 const int turbidityPin = 1; // Turbidity sensor analog pin (ADC)
12 const int levelPins[] = {3, 4, 5}; // Digital pins for level sensing via conductivity (3 levels)
13 const int numLevels = sizeof(levelPins) / sizeof(int);
14 const int tdsPin = 6; // TDS sensor analog pin (GravityTDS on GPIO6)
15 const int pumpPin = 33; // Pump control
16
17 // TDS sensor instance
18 GravityTDS gravityTds;
19
20 // Thresholds
21 const int tdsThreshold = 500; // ppm
22 const int turbThreshold = 100; // NTU
23
24 // Global variables
25 float tankSize = 1000.0; // Tank size in liters
26 float tankLevelPercent = 0.0; // From sensors
27 float tdsValue = 0.0; // TDS in ppm (float for library)
28 float tempC = 25.0; // Temperature in C (simulated)
29 float turbidityNTU = 0.0; // Turbidity
30 bool pumpOn = false;
31 bool highAlert = false;
32 String homeName = "Home1";
33 String alertMsg = ""; // For level-based alerts
  
```

Fig. 3: Code snippet configuring the Wi-Fi credentials

## TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to  
<https://sourcecode.electronicsforu.com>  
 to access and download source code  
 and other resources for this project

## Software

The GravityTDS.h library must be downloaded and added to the Arduino IDE through the Library Manager. After installation, the Wi-Fi SSID and password for network access are configured. The IndusBoard Coin supports STA (station) mode, AP (access point) mode, or both simultaneously. Example code is available for STA-only operation as well as combined STA + AP functionality.